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Problem 1

Solution

The last product contains $\overline{w} \cdot w = 0$, so that whole term is always 0.

Thus:

$$g = \overline{x}\overline{y}ztwq + \overline{x}yztw\overline{q}$$

Each product term is true for exactly one assignment:

- $\overline{x}\overline{y}ztwq = 1$: 1 row
- $\overline{x}yztw\overline{q} = 1$: 1 row

So $g = 1$ for 2 rows. With 6 variables, there are $2^6 = 64$ total rows.

Therefore, the answer is $64 - 2 = 2$

Problem 2

Solution

A, B, and C.
Let $M = Monkey$ and $D = Dolphin$

A: gives pure M, D formulas for xy and $\bar{x}$. Since AND, NOT is functionally complete, M, D is complete.

B: gives pure M, D formulas for $x + y$ and $\bar{x}$. Since OR, NOT is functionally complete, M, D is complete.

C: gives a pure M, D formula for $x \uparrow y$ (NAND). Since $NAND$ alone is functionally complete, M, D is complete.

D: gives only AND and OR, and AND, OR is not functionally complete.

E: expresses M, D in terms of NAND, which is the reverse direction. That does not prove NAND can be built from M, D.

F: expresses M, D using standard gates, also the reverse direction.

G: the second line gives $\bar{x}$, but the first line

$$xy = (xMy)(xDy)$$

still uses ordinary AND on the right side, so it is circular and does not give a pure M, D expression for AND.

Problem 3

Solution

The rows with output 0 are:

$(x, y, z) = (0, 1, 1)$ gives clause $(x \vee \bar{y} \vee \bar{z})$
$(x, y, z) = (1, 0, 1)$ gives clause $(\bar{x} \vee y \vee \bar{z})$
$(x, y, z) = (1, 1, 0)$ gives clause $(\bar{x} \vee \bar{y} \vee z)$

Therefore the CNF form is:

$f(x, y, z) = (x \vee \bar{y} \vee \bar{z})(\bar{x} \vee y \vee \bar{z})(\bar{x} \vee \bar{y} \vee z)$

Problem 4

Solution

A Boolean expression for the rules is:

$$E = (F_1+F_2+F_3)(\overline{F_1}+\overline{F_2})(\overline{F_1}+\overline{F_3})(\overline{F_2}+\overline{F_3})\cdot(A_1+A_2+A_3)(\overline{A_1}+\overline{A_2})(\overline{A_1}+\overline{A_3})(\overline{A_2}+\overline{A_3})\cdot(G_1+G_2+G_3)(\overline{G_1}+\overline{G_2})(\overline{G_1}+\overline{G_3})(\overline{G_2}+\overline{G}$$